

PEXIT Analysis and Design of Multilevel Coded Modulation With SPC-Aided SC-LDPC Codes

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Abstract—In this letter, we propose a multilevel coded modulation (MLCM) scheme based on single parity check (SPC)-aided spatially coupled low-density parity-check (SC-LDPC) codes to enhance spectral efficiency. A two-level MLCM framework is designed, in which the SC-LDPC component codes are compatible with the fifth-generation new radio (5G-NR) LDPC codes, and the message length allocations are optimized through the protograph-based extrinsic information transfer (PEXIT) analysis. Besides, an SPC constraint of message bits is introduced between different SC-LDPC codes to further lower the error floor. Simulation results demonstrate that the proposed MLCM scheme achieves significant gains over the conventional 5G-NR LDPC codes using bit-interleaved coded modulation, while also notably reducing decoding complexity.

Index Terms—Multilevel coded modulation, spatially-coupled low-density parity-check codes, protograph-based extrinsic information transfer, single parity check.

I. INTRODUCTION

WITH the rapid evolution towards the sixth-generation wireless communication systems, physical layer technologies are anticipated to undergo significant changes to meet the ever increasing demands for higher spectral efficiency and enhanced transmission reliability [1]. However, the bit-interleaved coded modulation (BICM) schemes employed in the fifth-generation new radio (5G-NR) systems with low-density parity-check (LDPC) codes, face critical challenges in balancing implementation complexity and ultra-high throughput, especially under high-order modulation scenarios [2].

Multilevel coded modulation (MLCM) addresses these challenges by decomposing the message into multiple bit levels and optimizing code protection per level [3]. Recent works have introduced probabilistic and geometric shaping techniques to further exploit shaping gain [4], [5], [6], as well as spatially-coupled (SC) codes to improve decoding thresholds and reduce the power consumption of channel decoding with high-order quadratic-amplitude modulation (QAM) [7]. In addition, several hierarchical and multi-rate coded modulation systems have explored bit-level rate trade-offs [8], [9], [10]. In [8], a source-constrained hierarchical modulation

architecture with LDPC codes is proposed, where inter-layer bit reuse and soft-information exchange enable flexible bit-level protection. However, most of these schemes involve high design complexity or limited scalability.

In this letter, we focus on the design of a two-level MLCM scheme based on single parity check (SPC)-aided SC-LDPC (SSC-LDPC) codes. Specifically, we employ the existing simplified error minimization progressive edge-growth (EMPEG) algorithm [11] to construct the SC-LDPC codes under heterogeneous lifting factors. Moreover, inspired by the inter-block cooperative coding framework in [12], an SPC constraint is introduced across different message blocks to reduce the error floor and enhance decoding performance [13]. Additionally, we utilize the protograph-based extrinsic information transfer (PEXIT) analysis [14] to optimize the message length allocations of the SC-LDPC component codes, which represent the code rate allocations across sub-blocks for performance enhancement. Simulation results demonstrate that the proposed scheme significantly outperforms the conventional 5G-NR BICM system in both decoding performance and computational complexity.

II. PROPOSED MLCM SCHEMES

A. Construction of SSC-LDPC Codes

To better balance the performance and complexity of MLCM schemes, we adopt SSC-LDPC codes due to the coupled structure leading to low complexity encoding and decoding. In this framework, each code block (CB) employs an SC-LDPC code, which ensures compatibility with the 5G-NR LDPC codes. Furthermore, to reduce the error floor, an SPC constraint on message bits is introduced across different CBs, forming a transport block (TB)-level coupling structure. Specifically, the base matrix $\mathbf{B}_{\text{SC}}$ for each CB is defined as

$$\mathbf{B}_{\text{SC}} = \begin{bmatrix} \mathbf{B}_{0,0} & & & & \\ \mathbf{B}_{0,1} & \mathbf{B}_{1,1} & & & \\ & \mathbf{B}_{1,2} & \ddots & & \\ & & \ddots & \mathbf{B}_{c-2,c-2} & \\ & & & \mathbf{B}_{c-2,c-1} & \mathbf{B}_{c-1,c-1} \end{bmatrix}, \quad (1)$$

where each diagonal sub-matrix $\mathbf{B}_{i,i}$, $i = 0, \dots, c-1$ represents the base matrix of the i -th component code, with size $m_i \times n_i$, while each sub-diagonal matrix $\mathbf{B}_{i,i+1}$, $i = 0, \dots, c-2$ with size $m_{i+1} \times n_i$ captures the coupling relationship between the adjacent components. All remaining entries are all-zero sub-matrices.

The corresponding exponent and parity-check matrices of the SC-LDPC codes are denoted by $\mathbf{P}_{\text{SC}}$ and $\mathbf{H}_{\text{SC}}$,

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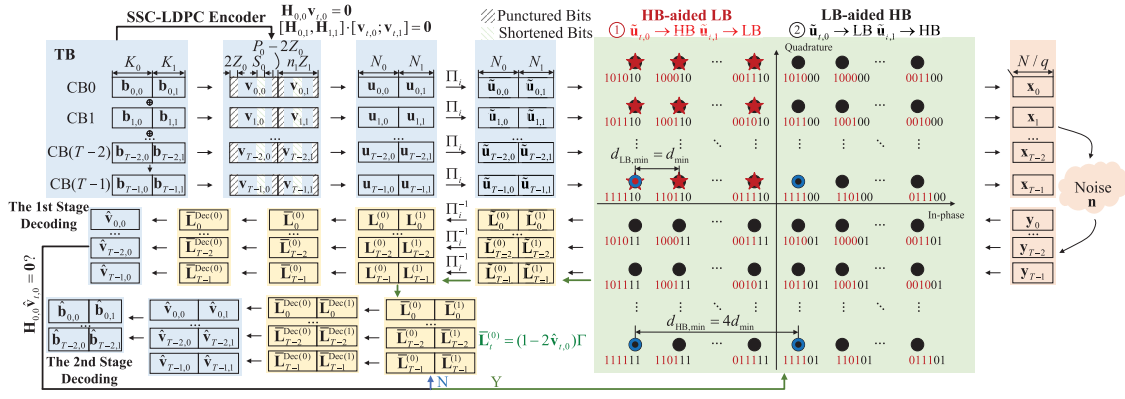


Fig. 1. The structure of the two-level MLCM scheme based on SSC-LDPC codes for 64-QAM.

respectively. To ensure compatibility with 5G standards, the diagonal sub-matrices $\mathbf{B}_{i,i}$, $\mathbf{P}_{i,i}$ and $\mathbf{H}_{i,i}$ are derived from the standardized base graphs and associated with the exponent matrices specified in 3GPP specifications [15]. Each element of $\mathbf{B}_{i,i}$ satisfies $(\mathbf{B}_{i,i})_{m,n} \in \{0, 1\}$. If $(\mathbf{B}_{i,i})_{m,n} = 0$, then $(\mathbf{P}_{i,i})_{m,n} = -1$, and the corresponding sub-matrix in $\mathbf{H}_{SC}$ is an all-zero matrix of size $Z_i \times Z_i$. If $(\mathbf{B}_{i,i})_{m,n} = 1$, then $(\mathbf{P}_{i,i})_{m,n} \in \{0, \dots, Z_i - 1\}$, indicating a circulant permutation of the identity matrix $\mathbf{I}_{Z_i}$ in $\mathbf{H}_{i,i}$.

For the coupling part, each sub-matrix $\mathbf{B}_{i,i+1}$ is set to $\mathbf{1}_{m_{i+1}, n_i}$, meaning all positions contain circulant permutation matrices. The corresponding exponent matrix $\mathbf{P}_{i,i+1}$ is optimized using the simplified EMPEG algorithm to reduce the error floor [11]. Each entry $(\mathbf{P}_{i,i+1})_{m,n} \in \{0, \dots, \max\{Z_i, Z_{i+1}\} - 1\}$ defines a circulant permutation sub-matrix of size $Z_{i+1} \times Z_i$ in $(\mathbf{H}_{i,i+1})_{m,n}$. The value specifies the starting index of the first '1' in the first column (resp., row) when $Z_i < Z_{i+1}$ (resp., $Z_i \geq Z_{i+1}$), with subsequent '1's inserted via cyclic shifting. A total of $\min\{Z_i, Z_{i+1}\}$ '1's are placed, with the final '1' appearing at index

$$((\mathbf{P}_{i,i+1})_{m,n} + \min\{Z_i, Z_{i+1}\} - 1) \bmod \max\{Z_i, Z_{i+1}\}. \quad (2)$$

Consequently, the code rate R_{SC} for each CB is denoted as

$$R_{SC} = \frac{K}{N} = \frac{\sum_{i=0}^{c-1} K_i}{\sum_{i=0}^{c-1} N_i} = \frac{\sum_{i=0}^{c-1} (n_i - m_i) Z_i - S_i}{\sum_{i=0}^{c-1} n_i Z_i - S_i - P_i}, \quad (3)$$

where K_i , S_i , N_i and P_i denote the numbers of message bits, shortened bits, coded bits and punctured bits for the i -th component code, respectively. K and N are the numbers of message and coded bits per CB, respectively.

To further strengthen the coupling among the T CBs within a TB, an SPC constraint of message bits is introduced. Let $\mathbf{b}_t = [\mathbf{b}_{t,0}, \dots, \mathbf{b}_{t,c-1}] \in \{0, 1\}^{K \times 1}$ denote the message vector of the t -th CB, with $\mathbf{b}_{t,i} \in \{0, 1\}^{K_i \times 1}$, $t = 0, \dots, T-2$, $i = 0, \dots, c-1$ being its i -th component code subvector. For $t = T-1$, the message vector $\mathbf{b}_{T-1}$ is constructed as the bit-wise XOR of the preceding $T-1$ CBs, i.e.,

$$\mathbf{b}_{T-1} = \mathbf{b}_0 \oplus \mathbf{b}_1 \oplus \dots \oplus \mathbf{b}_{T-2}. \quad (4)$$

This ensures that the combined message vectors satisfy an SPC constraint, thereby introducing redundancy that further enhances inter-block dependency and decoding robustness.

As a result, the number of independent message bits in a TB is reduced to $(T-1)K$. The sequence of original message bits and the last K XOR bits $\mathbf{b} = [\mathbf{b}_0, \dots, \mathbf{b}_{T-1}] \in \{0, 1\}^{TK \times 1}$ are encoded by T parallel encoders of our proposed SSC-LDPC codes to generate the encoded sequence $\mathbf{v} = [\mathbf{v}_{0,0}, \dots, \mathbf{v}_{T-1,c-1}]$. The encoding process of the SSC-LDPC codes is illustrated in Fig. 1. After puncturing, the transmitted vector is given by $\mathbf{u} = [\mathbf{u}_{0,0}, \dots, \mathbf{u}_{T-1,c-1}] \in \{0, 1\}^{TN \times 1}$. Thus, the code rate R_{SSC} of the TB is denoted as $R_{SSC} = (1 - \frac{1}{T}) R_{SC}$. As T increases, the code rate loss induced by the SPC constraint becomes almost negligible.

B. MLCM Schemes With SSC-LDPC Codes

We propose a two-level ($c = 2$) MLCM scheme based on SSC-LDPC codes for high-order QAM, as illustrated in Fig. 1 for the case of 64-QAM. Specifically, for Q -QAM, let $q = \log_2 Q$ and $q_0 + q_1 = q$, where $q_i, i = 0, 1$ represents the number of bits per constellation symbol assigned to the i -th component code of a CB. Note that the number of MLCM levels c is identical to the number of sub-matrices in a CB, ensuring consistency between coding and modulation. To balance the bit mapping between different modulation levels, we impose the constraint $N_0 q_1 = N_1 q_0$. Each vector $\mathbf{u}_{t,i}$ is passed through a row-column interleaver Π_i of size $q_i \times \frac{N_i}{q_i}$, which follows the interleaving structure specified in the 5G standard [16], and the interleaved output is denoted by $\tilde{\mathbf{u}}_{t,i}$.

During modulation, for instance in 64-QAM with $q_0 = 2$ or 4, the constellation is partitioned into two segments based on bit-level reliabilities. As shown in Fig. 1, the four red-labeled bits and the two black-labeled bits represent the low-reliability bits (LBs) and high-reliability bits (HBs), respectively. If the vector $\tilde{\mathbf{u}}_{t,0}$ belongs to the component code with higher code rate, it should be mapped to the HBs which correspond to larger minimum Euclidean distances $d_{HB,min} = 4 d_{min}$ in the possible partitioned constellation subset $\chi_i = \frac{d_{min}}{2} \{1 + 1j, -(\sqrt{Q} - 1) + 1j, -(\sqrt{Q} - 1) - (\sqrt{Q} - 1)j, 1 - (\sqrt{Q} - 1)j\}$ (the blue constellation points shown in Fig. 1), thereby providing better protection. Other possible subsets can be obtained by shifting this set along the real and/or imaginary axes. Here, j denotes the imaginary unit. In contrast, the vector $\tilde{\mathbf{u}}_{t,1}$, corresponding to the lower-rate component code, is mapped to the LBs, which have the minimum Euclidean distance $d_{LB,min} = d_{min}$. This mapping

corresponds to the “HB-aided LB” scheme, while swapping the two mappings yields the “LB-aided HB” scheme. The modulated signal vector is denoted as $\mathbf{x} = [\mathbf{x}_0, \dots, \mathbf{x}_{T-1}]$, which is then transmitted over an additive white Gaussian noise channel. The received signal $\mathbf{y} \in \mathbb{C}^{\frac{TN}{q} \times 1}$ is given by

$$\mathbf{y} = \mathbf{x} + \mathbf{n}, \quad (5)$$

where $\mathbf{n} \sim \mathcal{CN}(0, \sigma_n^2 \mathbf{I})$ is a complex Gaussian noise vector.

At the receiver, $\mathbf{y}$ is processed by a soft demodulator to compute the initial log-likelihood ratio (LLR) vector $\tilde{\mathbf{L}} = [\tilde{\mathbf{L}}_0, \dots, \tilde{\mathbf{L}}_{T-1}] \in \mathbb{R}^{TN \times 1}$, where each subvector is further split into $\tilde{\mathbf{L}}_t = [\tilde{\mathbf{L}}_t^{(0)}, \tilde{\mathbf{L}}_t^{(1)}] = [\tilde{L}_{t,0}, \dots, \tilde{L}_{t,N_0-1}, \tilde{L}_{t,N_0}, \dots, \tilde{L}_{t,N}]$. The LLR vector $\tilde{\mathbf{L}}_t^{(0)}$ is then passed through the deinterleaver Π_0^{-1} to recover the reordered LLR $\tilde{\mathbf{L}}_t^{(0)}$. Next, the punctured positions are filled with zeros to represent erasures, while the shortened bits are assigned with a large positive constant Γ to represent known zero-padding for $(-1)^{0/1}$ mapping. The resulting vector $\tilde{\mathbf{L}}_t^{(0)} \in \mathbb{R}^{n_0 Z_0 \times 1}$ is passed to an Approximate-Min* belief propagation (BP) decoder [17], which provides near-optimal performance with much lower complexity than full BP, to obtain the soft information and estimated encoded sequence $\tilde{\mathbf{L}}_t^{\text{Dec}(0)} \in \mathbb{R}^{n_0 Z_0 \times 1}$ and $\hat{\mathbf{v}}_{t,0}$, respectively. To avoid potential error propagation, as illustrated in Fig. 1, if the 1st stage decoding fails, i.e., $\mathbf{H}_{0,0} \cdot \hat{\mathbf{v}}_{t,0} \neq \mathbf{0}$, then $\tilde{\mathbf{L}}_t^{(0)}$ is updated utilizing the extrinsic information $\tilde{\mathbf{L}}_t^{e(0)} = \tilde{\mathbf{L}}_t^{\text{Dec}(0)} - \tilde{\mathbf{L}}_t^{(0)}$. Conversely, if $\mathbf{H}_{0,0} \cdot \hat{\mathbf{v}}_{t,0} = \mathbf{0}$, then each LLR is updated as

$$\tilde{L}_{t,n} = (1 - 2\hat{v}_{t,n}) \Gamma, n = 0, \dots, n_0 Z_0 - 1. \quad (6)$$

After re-puncturing, shortening, and re-interleaving, the updated vector $\tilde{\mathbf{L}}_t^{(0)}$ can assist the demodulation to obtain $\tilde{\mathbf{L}}_t^{(1)}$.

Similarly, $\tilde{\mathbf{L}}_t^{(1)}$ is deinterleaved to obtain $\tilde{\mathbf{L}}_t^{(1)}$, followed by the recovery of shortened and punctured bits to yield $\tilde{\mathbf{L}}_t^{(1)}$, and thus obtain $\tilde{\mathbf{L}}_t$. In the 2nd stage decoding, the parity-check matrix $[\mathbf{H}_{0,1}, \mathbf{H}_{1,1}]$ is used to produce the vector $\tilde{\mathbf{L}}_t^{\text{Dec}} = [\tilde{\mathbf{L}}_t^{\text{Dec}(0)}, \tilde{\mathbf{L}}_t^{\text{Dec}(1)}]$. If $\hat{\mathbf{v}}_{t,0}$ is successfully decoded in the 1st stage, the sub-matrix $\mathbf{H}_{0,1}$ can be excluded, and decoding is performed using only $\mathbf{H}_{1,1}$ with $\tilde{\mathbf{L}}_t^{(1)}$ as input, thereby reducing decoding complexity. Besides, although soft-information exchange across CBs is theoretically possible, its complexity becomes prohibitive for large T [13]. Hence, we focus on the typical case where only one CB fails and use the hard-decision outputs of other CBs to recover the erasures, reducing the error floor with much lower complexity.

III. PEXIT ANALYSIS FOR MLCM SCHEMES

The conventional PEXIT algorithm depends solely on the structure of the base matrix and is agnostic to the decoding strategy, making it unsuitable for MLCM schemes where decisions at one level affect others [14]. To address this issue, we propose a modified PEXIT algorithm inspired by bit error rate (BER) matching [18] and tailored for MLCM schemes with SC-LDPC codes. This enables the optimization of the message length allocations, i.e., the ratio K_0/K_1 , thereby improving the decoding threshold.

We begin by computing the initial BER $P_e^q(s)$, $q = 0, \dots, q-1$ under the bit labeling of a Q -QAM constellation. Here, $s = \frac{q d_{\min}^2}{4} \frac{E_b}{N_0}$, where E_b/N_0 denotes the ratio of energy per bit to the noise power spectral density. As shown in Fig. 1, for $q = 0$, if the real part of the received symbol satisfies $\Re\{y_{t,n}\} \in \mathcal{O}_{q=0}, n = 0, \dots, \frac{N}{q} - 1$, where

$$\mathcal{O}_{q=0} = \{r \in \mathbb{R} \mid r < -4d_{\min} \text{ or } 0 < r < 4d_{\min}\}, \quad (7)$$

then the first labeling bit is determined as 1; otherwise, it is 0. The corresponding BER is given by

$$P_e^0(s) = \frac{1}{8} \left(3\text{erfc}(\sqrt{s}) + 3\text{erfc}(3\sqrt{s}) - 2\text{erfc}(5\sqrt{s}) - 2\text{erfc}(7\sqrt{s}) + \text{erfc}(9\sqrt{s}) + \text{erfc}(11\sqrt{s}) \right), \quad (8)$$

where $\text{erfc}(\cdot)$ represents the complementary error function. The same procedure can be applied to compute $P_e^q(s)$ for other bit positions. Next, the equivalent signal-to-noise ratio and mutual information (MI) for the q -th bit are denoted as

$$s_{\text{eq}}^q = (\text{erfc}^{-1}(2P_e^q(s)))^2, I_{\text{ch}}^q = J\left(\sqrt{8s_{\text{eq}}^q}\right), \quad (9)$$

where $J(\cdot)$ is the standard MI transfer function [19].

As PEXIT analysis is performed per CB, we omit t for simplicity. Let $\tilde{\mathbf{I}}_{\text{ch}} = [\tilde{\mathbf{I}}_{\text{ch}}^{(0)}, \tilde{\mathbf{I}}_{\text{ch}}^{(1)}]$ and $\mathbf{I}_{\text{ch}} = [\mathbf{I}_{\text{ch}}^{(0)}, \mathbf{I}_{\text{ch}}^{(1)}] \in \mathbb{R}^{N \times 1}$ represent the MI vectors before and after deinterleaving, respectively. Each element in $\tilde{\mathbf{I}}_{\text{ch}}$ is selected from I_{ch}^q according to the modulation bit mapping. For example, if “LB-aided HB” is utilized, then $\tilde{\mathbf{I}}_{\text{ch}}^{(0)} = [I_{\text{ch}}^0, \dots, I_{\text{ch}}^{q_1-1}, I_{\text{ch}}^0, \dots, I_{\text{ch}}^{q_1-1}, \dots]$ and $\tilde{\mathbf{I}}_{\text{ch}}^{(1)} = [I_{\text{ch}}^{q_1}, \dots, I_{\text{ch}}^{q-1}, I_{\text{ch}}^{q_1}, \dots, I_{\text{ch}}^{q-1}, \dots]$. Then, we recover the channel MI vector $\tilde{\mathbf{I}}_{\text{ch}} = [\tilde{\mathbf{I}}_{\text{ch}}^{(0)}, \tilde{\mathbf{I}}_{\text{ch}}^{(1)}] = [\tilde{I}_{\text{ch},0}, \dots, \tilde{I}_{\text{ch},n_0 Z_0 + n_1 Z_1 - 1}]$ corresponding to the encoded sequence, where punctured and shortened bits are set to 0 and 1, respectively. Additionally, if a component code satisfies $K_i/N_i = 1$ and $P_e^q(s) < 10^{-5}$, we assume perfect reliability and set $\tilde{\mathbf{I}}_{\text{ch}}^{(i)} = \mathbf{1}$. Thus, the n -th element of the channel MI vector $\tilde{\mathbf{I}}_{\text{pch}} = [\tilde{\mathbf{I}}_{\text{pch}}^{(0)}, \tilde{\mathbf{I}}_{\text{pch}}^{(1)}] = [\tilde{I}_{\text{pch},0}, \dots, \tilde{I}_{\text{pch},n_0 + n_1 - 1}]$ of the base matrix can be denoted as

$$\tilde{I}_{\text{pch},n} = \begin{cases} \frac{1}{Z_i} \sum_{z=0}^{Z_i-1} \tilde{I}_{\text{ch},n Z_i + z}, i = 0 \\ \frac{1}{Z_i} \sum_{z=0}^{Z_i-1} \tilde{I}_{\text{ch},n_0 Z_0 + (n-n_0) Z_i + z}, i = 1 \end{cases}. \quad (10)$$

We then perform standard PEXIT analysis using $\tilde{\mathbf{I}}_{\text{pch}}^{(0)}$ and $\mathbf{B}_{0,0}$ [14]. If the posteriori MI satisfies $\mathbf{I}_{\text{APP}}^{(0)} = \mathbf{1}$, the 1st stage decoding is considered successful, and its output assists the 2nd stage decoding by setting $\tilde{\mathbf{I}}_{\text{ch}}^{(0)} = \mathbf{I}_{\text{ch}}^{(0)} = \mathbf{1}_{N_0 \times 1}$, $\tilde{\mathbf{I}}_{\text{ch}}^{(0)} = \mathbf{1}_{n_0 Z_0 \times 1}$, and $P_e^q(s) = 0$ for $q = 0, \dots, q_0 - 1$.

Then, if $\hat{\mathbf{v}}_{t,0}$ is error-free, the 2nd PEXIT analysis recomputes $P_e^q(s)$, $q = q_0, \dots, q-1$ using a priori information. For example, under the “LB-aided HB” scheme with $(q_0, q_1) = (4, 2)$, the updated BER for $q = 4$ is $P_e^4(s) = \frac{1}{2} \text{erfc}(4\sqrt{s})$. If the 1st stage decoding fails, the vector $\tilde{\mathbf{I}}_{\text{pch}}^{(0)}$ is updated through extrinsic MI of the decoder. Then, the subsequent

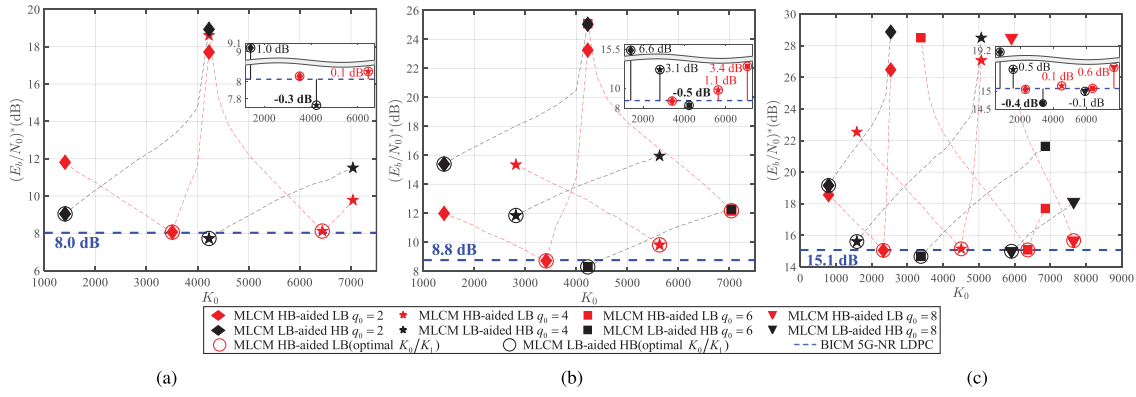


Fig. 2. PEXIT analysis of the proposed MLCM schemes ($K = 8448$). (a) 64-QAM ($R_{SC} = \frac{2}{3}$). (b) 256-QAM ($R_{SC} = \frac{1}{2}$). (c) 1024-QAM ($R_{SC} = \frac{2}{3}$).

steps follow the same procedure as in the 1st PEXIT analysis, except that the updated vector $\bar{\mathbf{I}}_{\text{pch}}$ is used as the input to the PEXIT analysis of $[\mathbf{B}_{0,1}, \mathbf{B}_{1,1}]$. If the condition $\mathbf{I}_{\text{APP}} = \mathbf{1}$ is met, successful decoding is declared at corresponding E_b/N_0 . Thus, the decoding threshold $(E_b/N_0)^*$ can be denoted as

$$(E_b/N_0)^* = \min \{E_b/N_0 \mid \mathbf{I}_{\text{APP}} = \mathbf{1}\}. \quad (11)$$

Finally, by comparing the thresholds across different (q_0, q_1) and modulation schemes, the optimal ratio K_0/K_1 that minimizes $(E_b/N_0)^*$ can be identified, providing a theoretical guideline for the design of MLCM schemes. Since the SPC constraint in SSC-LDPC codes only performs hard-decision recovery and the rate loss becomes negligible for large T , the same PEXIT analysis remains valid.

IV. THRESHOLD ANALYSIS AND SIMULATION RESULTS

A. PEXIT-Based Threshold Analysis for MLCM Schemes

Fig. 2 presents the decoding thresholds $(E_b/N_0)^*$ obtained through PEXIT analysis for different (q_0, q_1) and modulation schemes versus K_0 under Q -QAM. For each case, since $(E_b/N_0)^*$ varies with K_0 in a monotonic or convex manner, we only mark the minimum, maximum, and optimal points, while the intermediate points are also plotted on the curve without additional markers to avoid clutter, which is sufficient to illustrate the overall trend and identify the optimal K_0/K_1 ratio. The message length of each CB is fixed to $K = 8448$, with $K_1 = K - K_0$. For clarity, each inset shows the optimal thresholds and corresponding K_0 values, while also annotating the performance gains relative to the thresholds of 5G-NR LDPC coded BICM systems.

In Fig. 2(a), for 64-QAM ($R_{SC} = 2/3$), we consider two cases with $(q_0, q_1) = (2, 4)$ and $(4, 2)$, respectively. The “LB-aided HB” scheme with $(q_0, q_1) = (4, 2)$ achieves the lowest threshold of 7.7 dB at $K_0 = 4224$, i.e., $K_0/K_1 = 1.0$, which is 0.3 dB lower than that of the BICM system. In Fig. 2(b), for 256-QAM ($R_{SC} = 1/2$), it is shown that when $(q_0, q_1) = (6, 2)$, $(E_b/N_0)^* = 8.3$ dB is minimal, achieved at $K_0 = 4224$, i.e., $K_0/K_1 = 1.0$, for the “LB-aided HB” scheme, which is 0.5 dB lower than that of the BICM system. Besides, in Fig. 2(c), for 1024-QAM ($R_{SC} = 2/3$), the “LB-aided HB” schemes with $(q_0, q_1) = (6, 4)$ at $K_0 = 3380$ and $(q_0, q_1) = (8, 2)$ at $K_0 = 5914$ achieve thresholds that are 0.4 dB and 0.1 dB lower than that of the BICM system, respectively. These results further confirm

that the “LB-aided HB” schemes consistently achieve superior threshold performances across various (q_0, q_1) .

B. Simulation Results of Different MLCM Schemes

In this subsection, we compare BLER performance per CB (excluding the $(T - 1)$ -th CB containing only parity bits) of the MLCM schemes under different (q_0, q_1) and the optimal K_0/K_1 ratios obtained in Section IV-A. Besides, we examine the impact of introducing the SPC constraints across CBs on the decoding performance. The 5G-NR LDPC coded BICM system is included as a benchmark with the same code rate and block length. Simulation parameters are fixed as $K = 8448$ and $T = 100$, and the maximum number of decoding iterations is set to 10. The vertical dashed lines indicate the numerical thresholds of different schemes.

Across all modulation orders Q and code rates R_{SC} , the “LB-aided HB” schemes consistently yield superior performance. In Fig. 3(a), for 64-QAM, the configuration $(q_0, q_1) = (4, 2)$ achieves about 0.2 dB gain over BICM system at $\text{BLER} = 10^{-3}$ when the SPC constraints are applied. Moreover, the SPC constraints significantly improve performance in the high-SNR region ($E_b/N_0 > 8.5$ dB). In Fig. 3(b), the 256-QAM case using $(q_0, q_1) = (6, 2)$ with the SPC constraints shows approximately 0.5 dB gain compared to the BICM system. In Fig. 3(c), the “LB-aided HB” schemes with $(q_0, q_1) = (6, 4)$ and $(8, 2)$ outperform the BICM system by 0.5 dB and 0.2 dB, respectively. These results verify the accuracy of the PEXIT-guided optimization and the benefit of SSC-LDPC codes in lowering the error floor, albeit with slight performance degradation at low SNRs due to code rate loss.

C. Decoding Complexity Analysis

Since the majority of computational complexity arises from check-node updates, we evaluate the number of “ $\boxplus$ ” operations required in the Approximate-Min* BP decoding [17] of an SC-LDPC code (each CB) to quantify the decoding complexity. Specifically, each check-node update involves $(d_{c_{i,m}} - 1)$ “ $\boxplus$ ” operations, where $d_{c_{0,m}}$ (resp., $d_{c_{1,m}}$) denotes the degree of the m -th check node for the matrix $\mathbf{H}_{0,0}$ (resp., $[\mathbf{H}_{0,1}, \mathbf{H}_{1,1}]$). Hence, the overall complexity is approximated as

$$N_{\boxplus} = \sum_{i=0}^{c-1} T_{\text{iter},i} \sum_{m=0}^{m_i Z_i - 1} (d_{c_{i,m}} - 1), \quad (12)$$

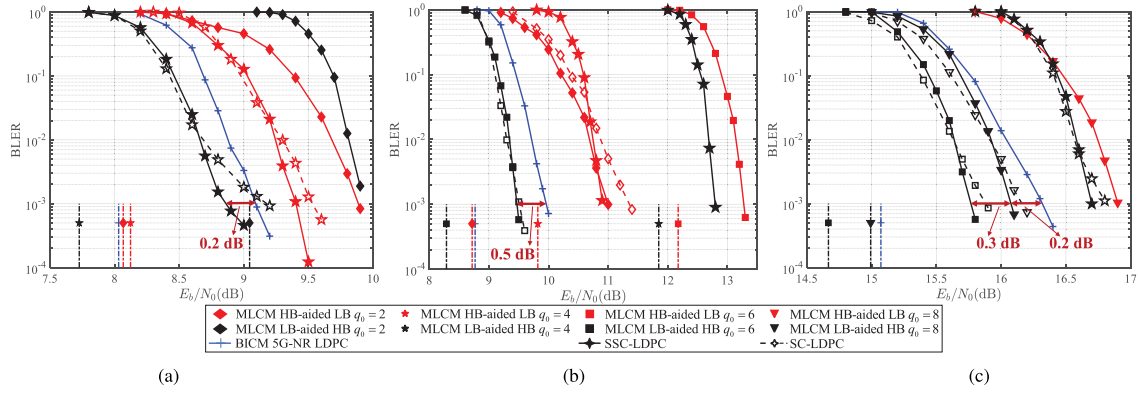


Fig. 3. BLER versus E_b/N_0 (dB) for MLCM and BICM schemes ($K = 8448, T = 100$). (a) 64-QAM ($R_{SSC} = 0.66$). (b) 256-QAM ($R_{SSC} = 0.50$). (c) 1024-QAM ($R_{SSC} = 0.66$).

TABLE I

NUMBER OF “田” OPERATIONS REQUIRED $N_{\text{田}} (\times 10^5)$ FOR OPTIMAL MLCM AND BICM SCHEMES AT BLER = 10^{-3}

Scheme	q_0	64-QAM	256-QAM	1024-QAM
BICM	—	4.7	6.7	4.6
MLCM	2	4.1	5.3	3.4
MLCM	4	5.3	6.1	3.9
HB-aided LB	6	—	5.3	4.1
HB-aided LB	8	—	—	5.0
MLCM	2	2.9	3.4	2.5
MLCM	4	3.1 (↓ 34.0%)	4.6	2.8
LB-aided HB	6	—	4.8 (↓ 28.4%)	3.1 (↓ 32.6%)
LB-aided HB	8	—	—	4.0

where $T_{\text{iter},i}$ is the average numbers of decoding iterations for the i -th component code of a CB.

Table I compares the decoding complexity of the optimal MLCM schemes with the SPC constraints at BLER = 10^{-3} , with parameters set as in Section IV-B. It is obvious that most MLCM schemes require fewer “田” operations than the BICM systems, and the “LB-aided HB” schemes consistently exhibit the lowest complexity, as the enhanced priori information at the HB layer allows faster convergence of the 2nd stage decoding. In particular, for $Q = 64, 256$ and 1024 , the “LB-aided HB” schemes with $q_0 = 4, 6$ and 6 achieve 34.0%, 28.4% and 32.6% reductions compared to corresponding BICM baselines, respectively, demonstrating the effectiveness of the proposed schemes under high-order modulation.

V. CONCLUSION

This letter has proposed a two-level MLCM scheme based on SSC-LDPC codes under high-order modulation, which can be readily extended to multilevel scenarios. A modified PEXIT algorithm, tailored to the MLCM framework, is employed to optimize the message length allocations of the SC-LDPC component codes, thereby minimizing the decoding threshold. In addition, an SPC constraint of message bits is introduced to lower the error floor. Simulation results demonstrate that the proposed scheme outperforms the 5G-NR BICM system in both decoding performance and computational complexity.

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